

Benchmarking Aggregated Results

Method	Comp. Ratio	Enc (ms)	Dec (ms)	PSNR (dB)	Peak Mem (MB)
GZIP (Raw .nii)	5.26	641.1	130.26	inf	85.91
Zstd (Raw .nii)	5.32	82.48	26.87	inf	34.19
NumPy+Zstd Level 3	5.32	148.47	63.31	inf	106.56
KMRI (8-bit, ROI Lvl 5)	21.11	1121.21	285.43	39.87	217.83
KMRI (10-bit, ROI Lvl 5)	14.43	1423.92	353.39	39.88	217.83
KMRI (12-bit, ROI Lvl 5)	11.9	1219.15	318.32	39.88	217.83
GZIP (Raw .nii)	5.61	460.98	32.1	inf	25.57
Zstd (Raw .nii)	5.58	11.32	1.98	inf	6.14
NumPy+Zstd Level 3	5.65	184.71	62.43	inf	106.56
KMRI (8-bit, ROI Lvl 5)	27.84	1608.05	517.89	43.58	216.24
KMRI (10-bit, ROI Lvl 5)	16.72	1334.0	360.08	43.59	216.24
KMRI (12-bit, ROI Lvl 5)	13.02	1503.26	430.09	43.59	216.24
GZIP (Raw .nii)	5.36	349.57	23.38	inf	26.13
Zstd (Raw .nii)	5.34	8.19	1.71	inf	6.41
NumPy+Zstd Level 3	5.39	175.49	60.99	inf	106.56
KMRI (8-bit, ROI Lvl 5)	26.17	1358.48	368.81	48.04	217.55
KMRI (10-bit, ROI Lvl 5)	15.81	1223.41	250.3	48.09	217.55
KMRI (12-bit, ROI Lvl 5)	12.25	1218.34	293.2	48.09	217.55
GZIP (Raw .nii)	5.35	392.35	25.0	inf	26.13
Zstd (Raw .nii)	5.34	9.65	2.1	inf	6.41
NumPy+Zstd Level 3	5.38	179.65	55.59	inf	106.56
KMRI (8-bit, ROI Lvl 5)	22.31	1314.87	292.59	54.72	216.58
KMRI (10-bit, ROI Lvl 5)	14.85	1243.6	278.33	54.77	216.58
KMRI (12-bit, ROI Lvl 5)	12.1	1179.55	254.33	54.78	216.58
GZIP (Raw .nii)	5.38	357.39	24.09	inf	26.06
Zstd (Raw .nii)	5.37	15.74	2.0	inf	6.38
NumPy+Zstd Level 3	5.41	181.49	56.91	inf	106.56
KMRI (8-bit, ROI Lvl 5)	21.76	1137.95	233.91	49.87	217.25
KMRI (10-bit, ROI Lvl 5)	14.35	1134.58	260.09	49.91	217.25
KMRI (12-bit, ROI Lvl 5)	12.0	1209.6	257.26	49.91	217.25
GZIP (Raw .nii)	5.39	369.62	24.26	inf	26.04
Zstd (Raw .nii)	5.38	9.59	1.85	inf	6.37
NumPy+Zstd Level 3	5.42	159.43	52.05	inf	106.56
KMRI (8-bit, ROI Lvl 5)	26.68	1229.93	261.04	49.11	217.38
KMRI (10-bit, ROI Lvl 5)	16.01	1155.88	241.75	49.14	217.38
KMRI (12-bit, ROI Lvl 5)	12.3	1539.44	389.94	49.14	217.38
GZIP (Raw .nii)	5.41	438.73	27.92	inf	26.01
Zstd (Raw .nii)	5.39	12.14	2.42	inf	6.35
NumPy+Zstd Level 3	5.45	238.85	71.72	inf	106.56
KMRI (8-bit, ROI Lvl 5)	20.14	1261.32	245.57	42.41	217.56
KMRI (10-bit, ROI Lvl 5)	13.45	1225.53	258.18	42.42	217.56
KMRI (12-bit, ROI Lvl 5)	11.46	1233.03	305.1	42.42	217.56
GZIP (Raw .nii)	5.43	407.75	26.93	inf	25.96
Zstd (Raw .nii)	5.41	10.52	2.08	inf	6.33
NumPy+Zstd Level 3	5.47	181.87	55.42	inf	106.56
KMRI (8-bit, ROI Lvl 5)	20.71	1089.03	250.99	42.28	217.28
KMRI (10-bit, ROI Lvl 5)	13.92	1148.63	248.93	42.28	217.28
KMRI (12-bit, ROI Lvl 5)	11.72	1043.42	195.5	42.28	217.28
GZIP (Raw .nii)	5.53	354.57	24.66	inf	25.74
Zstd (Raw .nii)	5.51	8.98	1.68	inf	6.22
NumPy+Zstd Level 3	5.57	142.65	46.98	inf	106.56
KMRI (8-bit, ROI Lvl 5)	29.67	997.0	228.19	43.71	216.53
KMRI (10-bit, ROI Lvl 5)	17.16	1206.05	279.03	43.71	216.53
KMRI (12-bit, ROI Lvl 5)	12.86	1057.17	213.27	43.71	216.53
GZIP (Raw .nii)	5.55	308.31	21.78	inf	25.7
Zstd (Raw .nii)	5.53	8.14	1.64	inf	6.2
NumPy+Zstd Level 3	5.58	147.47	51.93	inf	106.56
KMRI (8-bit, ROI Lvl 5)	26.12	927.97	227.84	45.35	217.15
KMRI (10-bit, ROI Lvl 5)	16.35	996.85	205.63	45.37	217.15
KMRI (12-bit, ROI Lvl 5)	13.02	1065.43	210.44	45.37	217.15